

Recap Checkpoint 6 — Full Year 12 Cumulative (Pre-Mock): 1.1–1.4 + 2.1 + 2.2.1

Name: _____ Date: _____ Mark: _____ / 46

OCR H446 cumulative recap — mixes every topic taught so far. Show all working.

Questions

Q1 [3] (AO1). (1.1 — Processors) State **three** factors that affect the **performance** of a processor, and for each give a brief explanation of its effect.

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Q2 [4] (AO2). (1.4 — Number bases) Show all working.

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(a) Convert denary **181** to **8-bit binary**. [1]

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(b) Convert **181** to **hexadecimal**. [2]

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(c) Convert binary **1010 0110** to denary. [1]

Q3 [4] (AO2). (1.4 — Two's complement) Using **8-bit two's complement**, calculate $-40 + 25$.

(a) Show 40 and 25 in 8-bit binary. [1]

(b) Form the two's complement representation of -40 . [1]

(c) Perform the addition and give the final **denary** result. [2]

Q4 [3] (AO2). (1.4 — Boolean algebra / floating point)

(a) State why floating-point numbers are **normalised**. [1]

(b) Simplify $A \vee A \wedge B$ and name the law used. [2]

Q5 [4] (AO1). (1.3 — Networks)

(a) Explain the difference between **circuit switching** and **packet switching**. [2]

(b) State **two** items of data added to a packet's **header**. [2]

Q6 [3] (AO1). (1.2 — Operating systems)

(a) State what is meant by **scheduling** and name **one** scheduling algorithm. [2]

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(b) State the purpose of a **device driver**. [1]

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Q7 [4] (AO2). (2.1 / 2.2.1 — *Computational thinking + programming*)

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(a) State the difference between a **procedure** and a **function**. [1]

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(b) Explain the difference between **passing a parameter by value** and **by reference**. [2]

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(c) State **one** advantage of using **local** variables over **global** variables. [1]

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Q8 [5] (AO2). (1.4 — *Data structures trace*) A **stack** is initially empty. The following operations are carried out in order: `push(7)` , `push(2)` , `push(9)` , `pop()` , `push(4)` , `pop()` , `pop()` .

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(a) State the value returned by **each** of the three `pop()` calls, in order. [2]

(b) State the contents of the stack after all seven operations. [1]

(c) The same seven operations are repeated using a **queue** (`enqueue` in place of `push`, `dequeue` in place of `pop`). State the values returned by the three `dequeue()` calls, and name the access order each structure demonstrates. [2]

Q9 [4] (AO1/AO2). (2.1 — Computational thinking)

(a) State what is meant by **abstraction** and give **one** example of its use in software. [2]

(b) **Caching** is often described as an example of *thinking ahead*. State what caching is and give **one** benefit it provides. [2]

Q10 [6] (AO1/AO2). (1.4 — Boolean logic: adders)

(a) Complete the **Sum** and **Carry** columns of the truth table for a **half adder**. [2]

A	B	Sum	Carry
0	0		
0	1		
1	0		
1	1		

(b) Name the **logic gate** that produces the Sum output and the logic gate that produces the Carry output. [2]

(c) State **one** way a **full adder** differs from a half adder, and explain why this difference allows full adders to be chained together to add multi-bit binary numbers. [2]

Q11 [6] (AO3). (Synoptic extended response) A hospital is replacing a paper appointment system with networked software that stores patient data, lets staff search records, and sends automated reminders.

In an extended response, discuss the design considering: **(i)** how **abstraction and decomposition** help manage the build, **(ii)** an appropriate **data structure** for fast record lookup (justify your choice), and **(iii)** **one security** consideration for transmitting and storing patient data. **[6]**
